circuitRL: Automated Analog Circuit Sizing via Deep Reinforcement Learning

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Problem

- Integrated circuits power nearly every modern device–from smartphones and laptops to cars and home appliances.
- A critical part of IC development is **analog circuit sizing,** which involves choosing circuit parameters to meet a desired set of specifications.
- This process is time-consuming and requires an expert designer to iteratively:
  - select parameters for each circuit element
  - simulate the circuit using a SPICE simulator
  - evaluate results against design requirements
  - repeat until all specifications are met

We ask: can an RL agent automate this process?

Background

- Analog circuits are networks of electronic components: **resistors, transistors, capacitors,** and more.
- Each component has parameters (**width, length, resistance,** etc.) that determine circuit behavior, measured by specifications like **gain, bandwidth, and power.**
- We frame analog circuit sizing as a **Markov Decision Process** (MDP) by discretizing the parameter space into a finite array of physically realizable values, then defining:

State	current parameter values, circuit metrics, & target specifications
Action	increment, decrement, or hold each parameter
Transition	deterministic circuit simulation using NGspice software
Reward	distance from current metrics to target specs + bonus for meeting specs to a desired tolerance

- Additionally, we formulate a second **graph-based MDP** where the state information is restructured onto a graph (with circuit components as nodes, and wires as edges)

State	1) Graph Structure: The circuit topology encoded as an adjacency matrix (fixed per circuit), 2) Node features: each component's parameter values 3) Current specs and target specs

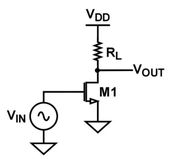
Methods

- We evaluate several RL approaches for our first MDP:
  - Shared-trunk PPO
    - With sequential parameter updates
    - With simultaneous parameter updates
    - With multiple step sizes
    - With continuous state space
  - Non shared-trunk PPO (regular PPO)
  - GRPO
- For our second graph MDP, we evaluate transfer learning across different circuit topologies with GNN PPO

Experiments

- One major category of analog circuits are **amplifiers** (amps), which amplify the voltage of a signal.
- We tested on two different types of amplifier circuits: a basic **common source amp** (with a single transistor), and a **two-stage operational-amp** (with three transistors)

Common source amp schematic

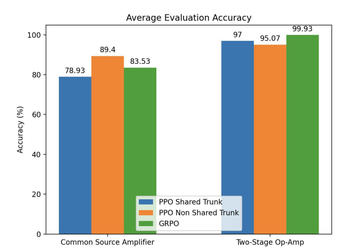


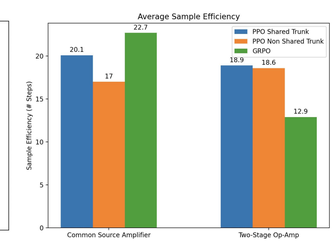
- The inputs to the model are the parameters of the components of the amp
  - Width and length of transistors, resistance of resistors
- The outputs of the model are the characteristics of the amp
  - How much it amplifies the voltage (gain) and over which frequencies (bandwidth)

- We conducted many experiments to try to create the best possible circuit solver with RL. We explored:
  - Number of unique specifications in training pool
  - Separate-trunk vs shared-trunk PPO
  - Joint parameter update vs sequential parameter update
  - Constant discrete step size vs multiple step sizes
  - Graph-based PPO model trained on one circuit (common source amp) and fine-tuned on another (cascaded common source amp, which is two common source amps connected together)

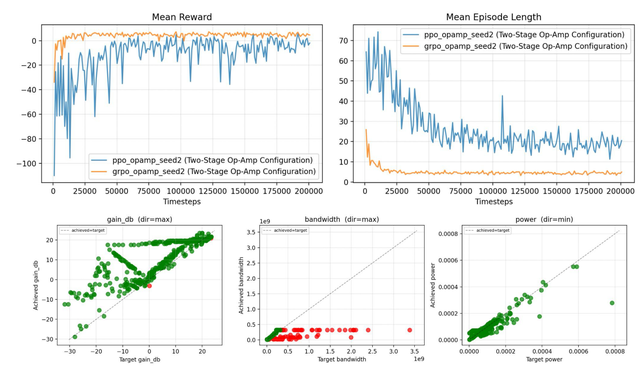
Analysis

- **Summary of different RL methods**
  - Accuracy = percent of randomly sampled target specifications that were met during evaluation
  - Sample efficiency = number of simulation steps needed to achieve target specification

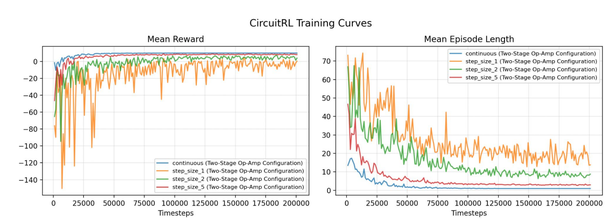




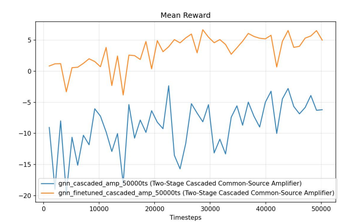
- **GRPO**
  - Comparable accuracy with PPO across both circuits
  - Significantly better sample efficiency for complex circuit with more parameters (two-stage op-amp)
  - Bandwidth is the hardest target specification to meet

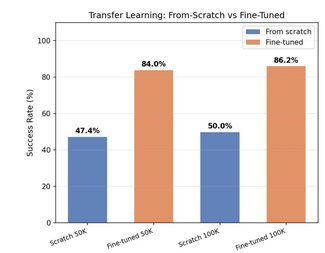


- **Discrete step sizes vs continuous action space**
  - Multiple step size options improved convergence for more complex circuit (two-stage op-amp), didn't seem to have major impact on simpler circuit
  - Continuous step sizes (which is basically the limit as step size options increase) seemed to be the most sample efficient and fastest to converge.



- **GNN PPO**
  - Fine-tuning a pretrained GNN dramatically outperforms training from scratch with GNN PPO on a new circuit, at the same simulation budget.





  - Right shows success rate on 500 cascaded amplifier targets. "From scratch" models were trained only on the cascaded amp; "fine-tuned" models were initialized from a CS-amp pretrained checkpoint, then trained on the cascaded amp.

Conclusions

- RL methods successfully automate circuit sizing to achieve around **~85-90%** of randomly sampled target specifications, across multiple circuit topologies.
- **Separate actor-critic networks improve training stability** and sample efficiency, particularly for circuits with many specifications and circuit parameters.
- **GRPO is competitive with PPO** in terms of accuracy, and avoids using a value function network. GRPO also gets significant sample efficiency improvements for circuits with many parameters.
- Joint parameter updates are significantly **more sample efficient than sequential updates**. For the latter, the increased simulation cost per episode makes learning infeasible.
- Multiple step sizes in the discrete parameter space (ie, [-2, -1, 0, +1, +2]) **outperforms single step sizes** ([-1, 0, +1]) and enable broad exploration in early training and finer steps for convergence. Continuous action space seemed to greatly improve performance and time to convergence.
- **Graph-based** RL approaches enable **transfer learning** across different circuit topologies.

References

Settaluri, K., Haj-Ali, A., Huang, Q., Hakhamaneshi, K., & Nikolic, B. (2020, March 1). *AutoCkt: Deep Reinforcement Learning of Analog Circuit Designs*. IEEE Xplore. https://doi.org/10.23919/DATE48585.2020.9116200

Wang, Hanrui, et al. (2020). *GCN-RL circuit designer: Transferable transistor sizing with graph neural networks and reinforcement learning.* 2020 57th ACM/IEEE Design Automation Conference (DAC), July 2020, pp. 1–6, https://doi.org/10.1109/dac18072.2020.9218757.